Enabling Community Collaboration and Open Science in Hydrology through Data and Model Sharing with CUAHSI HydroShare

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This research was supported by the Cooperative Institute for Research to Operations in Hydrology (CIROH) with funding under award NA22NWS4320003 from the NOAA Cooperative Institute Program. The statements, findings, conclusions, and recommendations are those of the authors and do not necessarily reflect the opinions of NOAA

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HydroShare is the CUAHSI Hydrologic Information System for sharing and computing with hydrologic data, models and code

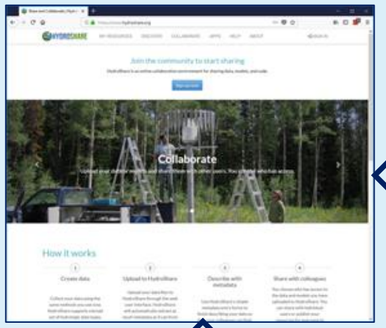
Collaboration, Reproducibility, Credit, Transparency

Server/Cloud platforms can be pre-configured to meet application needs

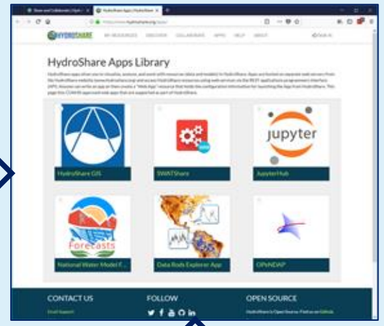
- **Sharing and publication of data (DOI)**
- **Social discovery and added value**
- **Model and workflow sharing**

Data and model repository Website

Gateway to Computing



API



OAuth

API

API

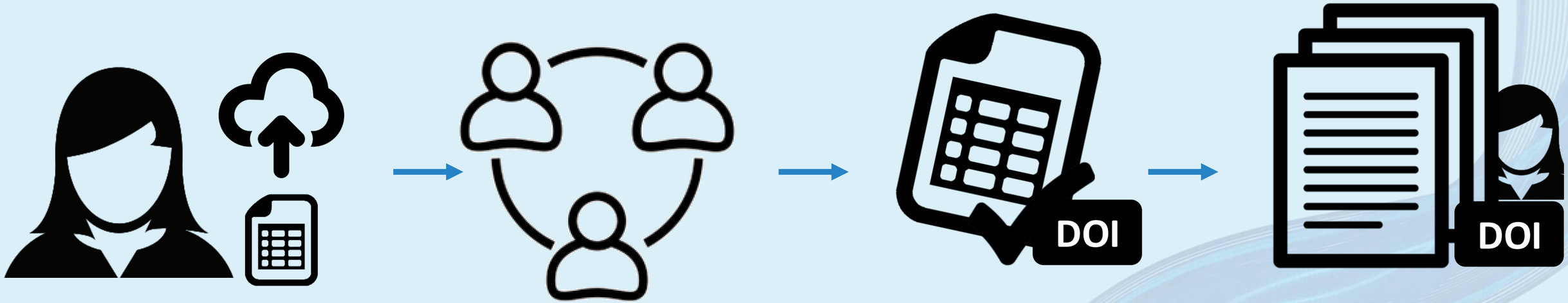
S3 Object Store

Standards based Resource Data Model

- **Reproducibility**
- **Model input data preparation**
- **Model execution**
- **Visualization and analysis (best of practice tools)**
- **Anyone can set up a server/platform and connect it with HydroShare using the HydroShare connectivity software stack (nbfetch, hsclient, hs_files)**

https://doi.org/10.1016/j.envsoft.2023.105902

HydroShare workflow



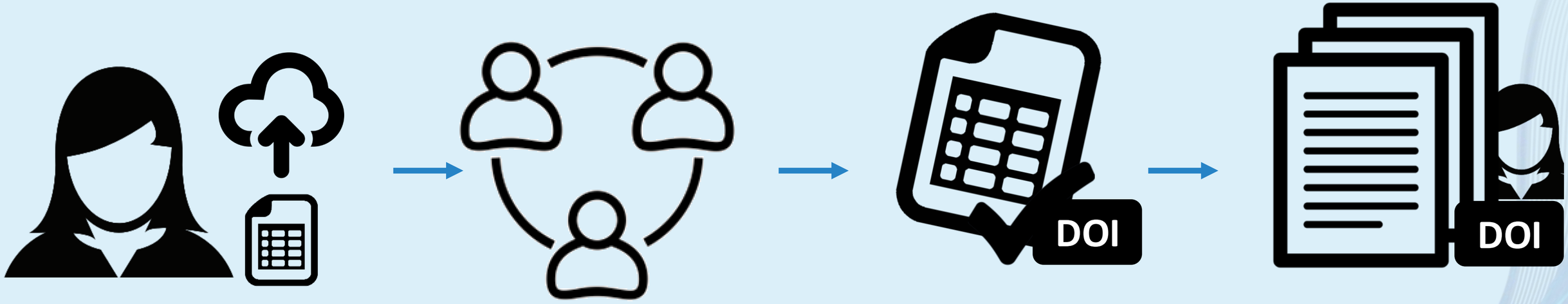
User creates dataset and uploads to HydroShare

Research team iterates collaboratively

Final version is published and assigned DOI

User cites dataset in paper

Example



Philippus creates dataset and uploads to HydroShare

Research team iterates collaboratively

Final version is published and assigned DOI

Philippus cites dataset in paper

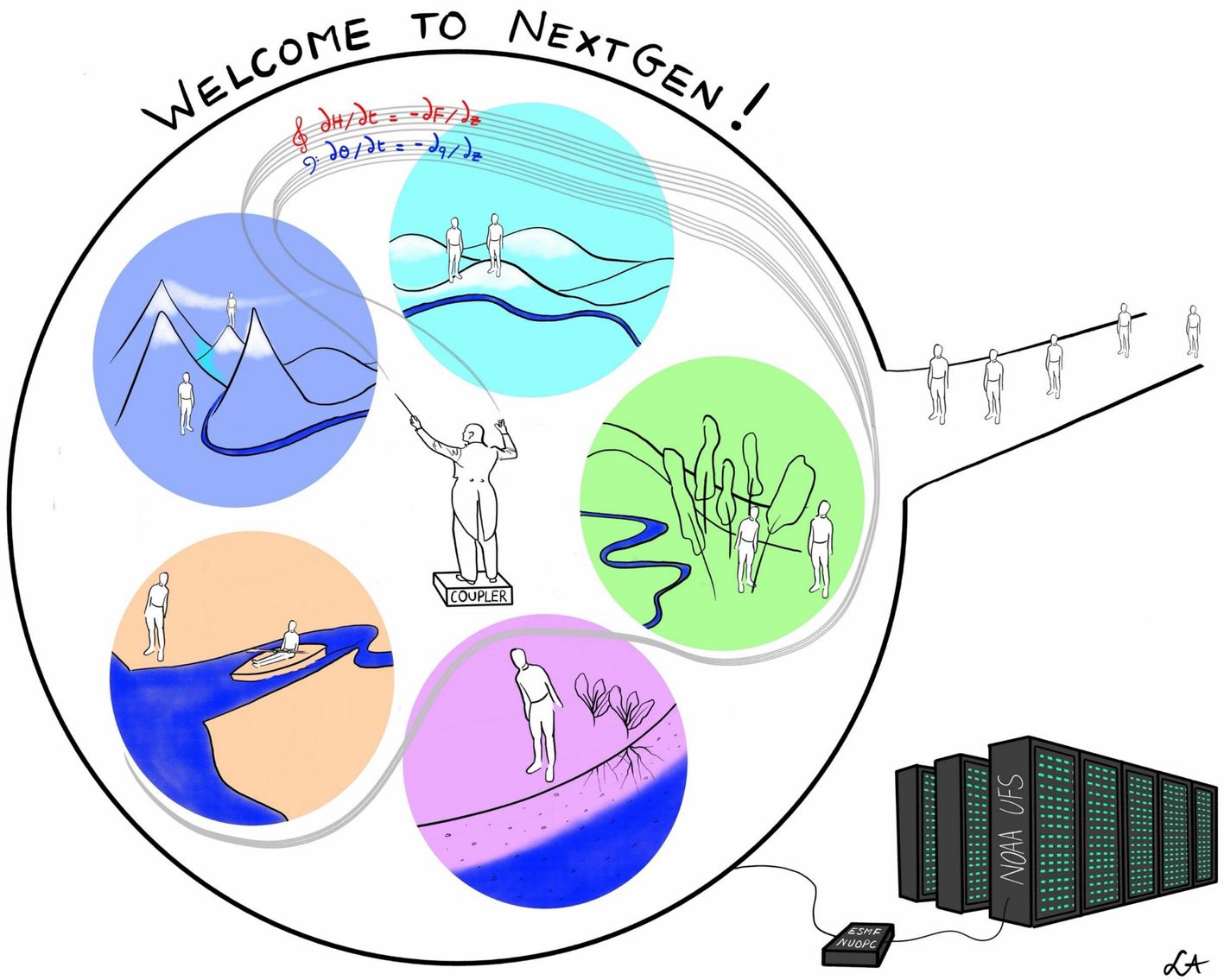
Philippus, D., C. R. Corona, K. Schneider, A. Rust, T. S. Hogue (2025). TempEst 2 Development Data: Observed Stream Temperature, Covariates, Performance Data, and Analysis Notebooks, HydroShare, https://doi.org/10.4211/hs.a8b243957f7946e388d10ab206990675

Philippus, D., C. R. Corona, K. Schneider, A. Rust and T. S. Hogue, (2025), "Satellite-based spatial-statistical modeling of daily stream water temperatures at the CONUS scale," Journal of Hydrology, 661: 133321, https://doi.org/10.1016/j.jhydrol.2025.133321.

Data and model availability

The TempEst 2 model source code, a pre-trained model, and a reproducible validation R Notebook are publicly available at https://github.com/mines-ciroh/TempEst2/ ↗. Model development data, the pre-trained model, and the compiled validation notebook results are also available through CUAHSI HydroShare at https://doi.org/10.4211/hs.a8b243957f7946e388d10ab206990675 ↗. The code may be reused, modified, extended, and redistributed under the terms of the GNU General Public License v3.

The NOAA/OWP NextGen Water Resources Modeling framework



- The rationale for NextGen is
  - A single model will not be best everywhere; and
  - A single model will not be best at everything.
- NextGen offers the flexibility to configure a **mosaic of alternative model formulations that represent different hydrologic processes** across different parts of the modeling domain.
- Models interoperate using the CSDMS Basic Model Interface (BMI) standard, over a Hydrofabric based on Hy_Features data model for representing hydrologic catchments (divides), flowpaths, and nexuses.

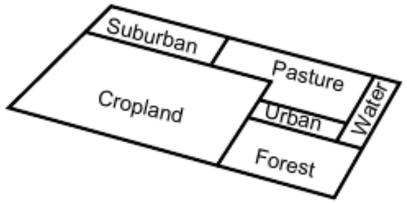
https://github.com/NOAA-OWP/NextGen-Info

Based on slide from Martyn Clark

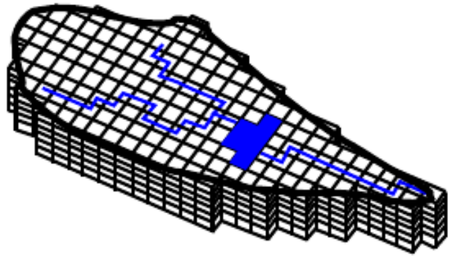
NextGen alternative algorithmic formulations contemplated

Watershed

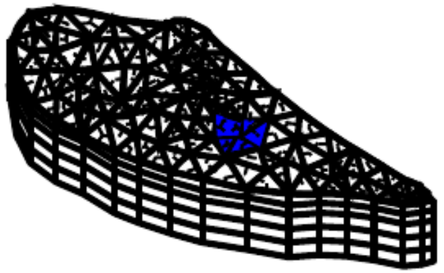
Hillslope



Tiled



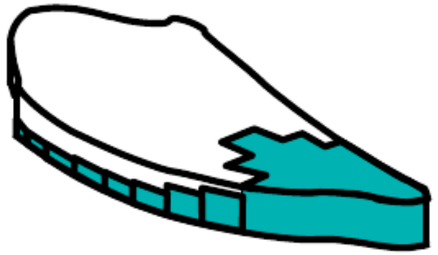
Structured



Unstructured/
Voronoi

Isochronous/
Stream Tube

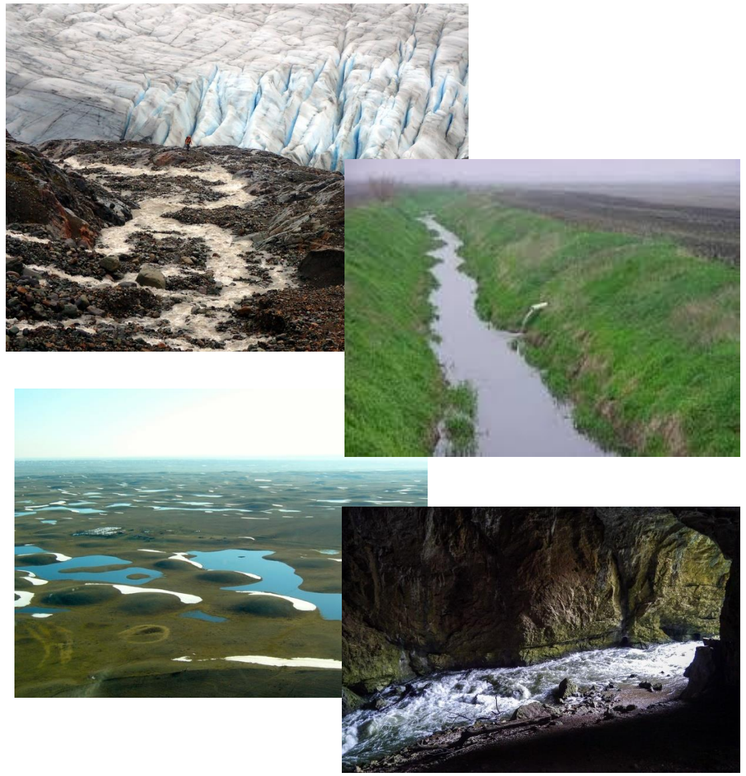
Conceptual



Topo. Wetness Index

ML

Select models to represent the variability of hydrologic processes across the modeling domain



Based on Slides from Fred Ogden and Martyn Clark

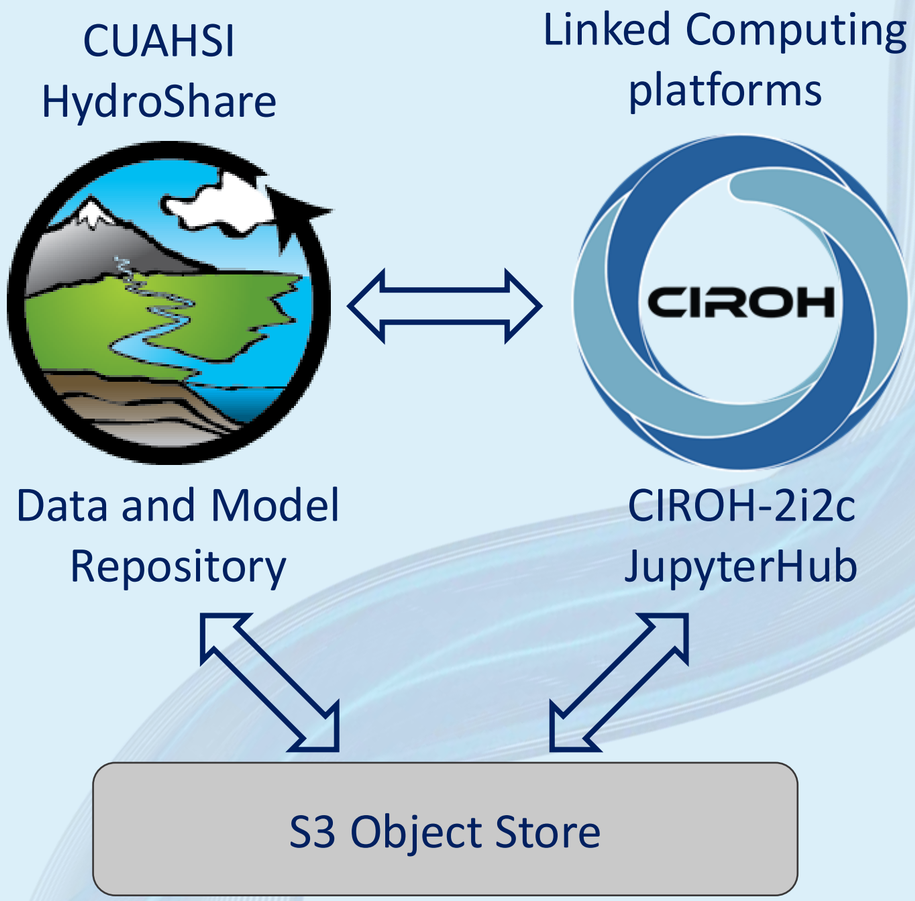
But it is complicated

Challenges

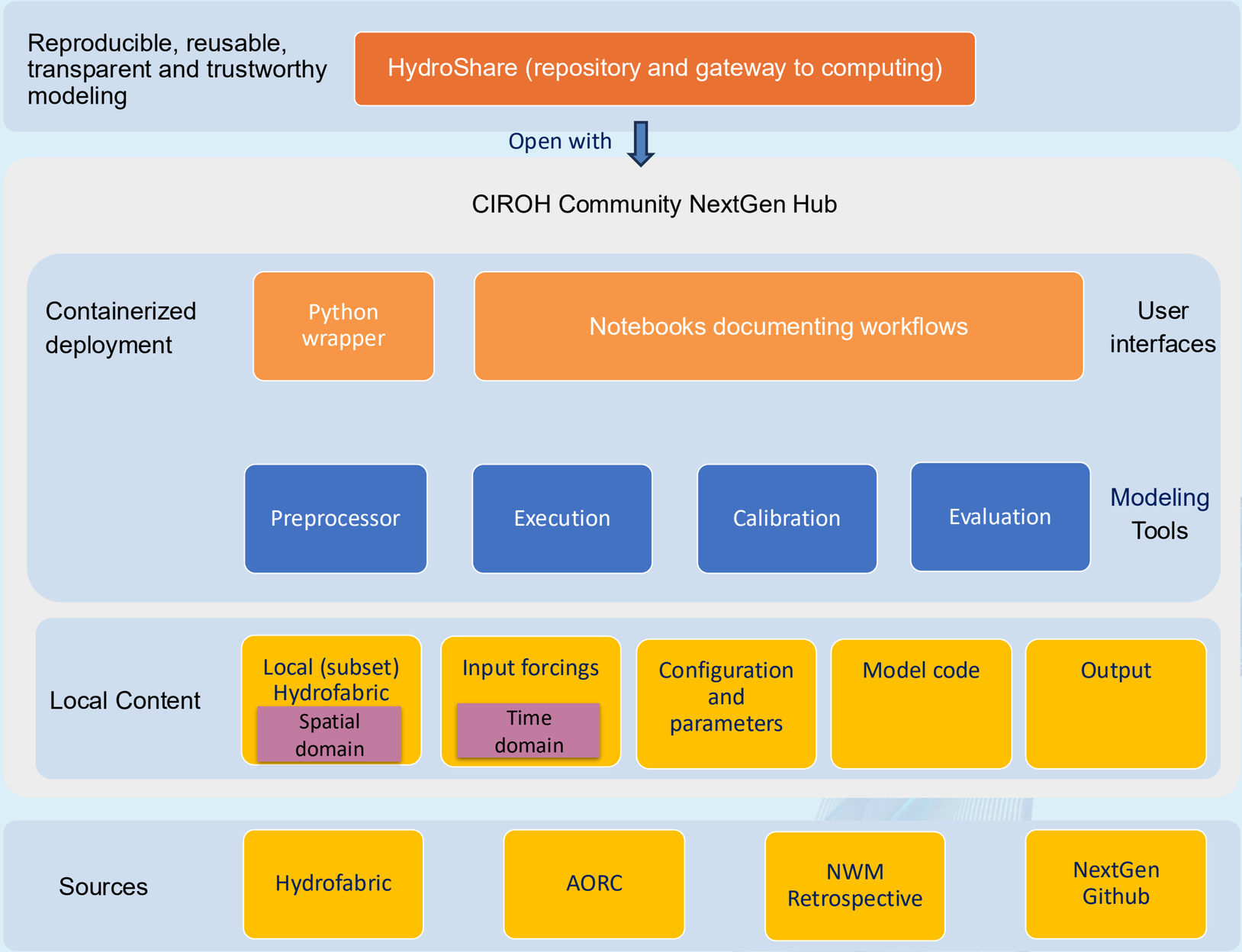
- Model complexity
- Technical demands of software dependencies, configuration and compatibility
- Input data preparation
- Model configuration

Towards a solution

- Cloud based cyberinfrastructure (HydroShare, S3 storage, CIROH-2i2c JupyterHub)
- CIROH-2i2c JupyterHub configured with
  - NextGen In A Box (NGIAB) box software
  - Data Preprocessing tools
  - Analysis and visualization tools
- Jupyter notebooks to document workflows
- An entry point for research to improve the National Water Model as it adopts the NextGen framework



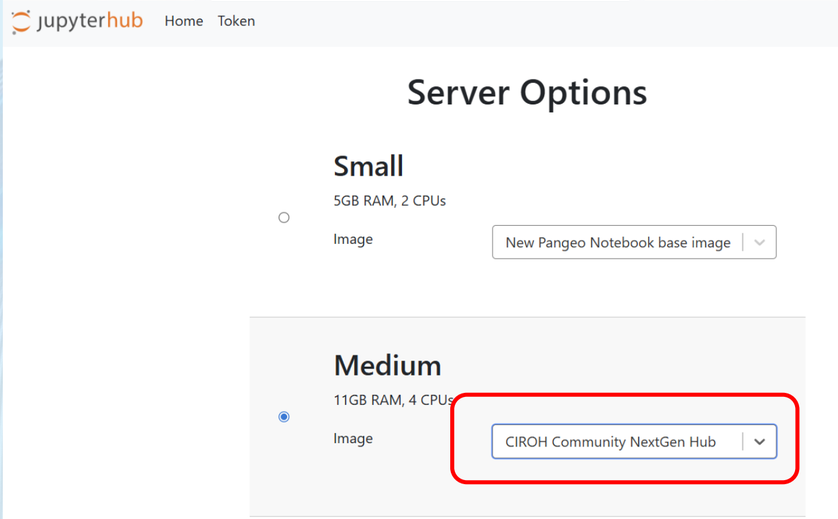
Architecture of containerized deployment of 2i2C JupyterHub with NextGen Software Ecosystem



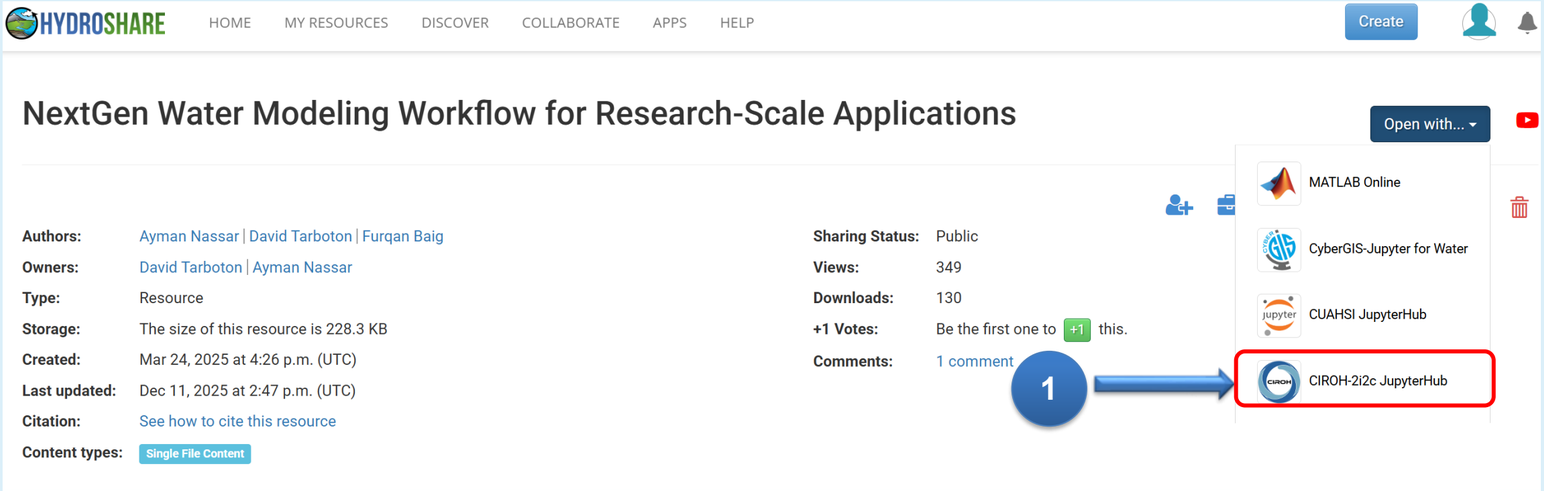
Software in CIROH Community NextGen Hub

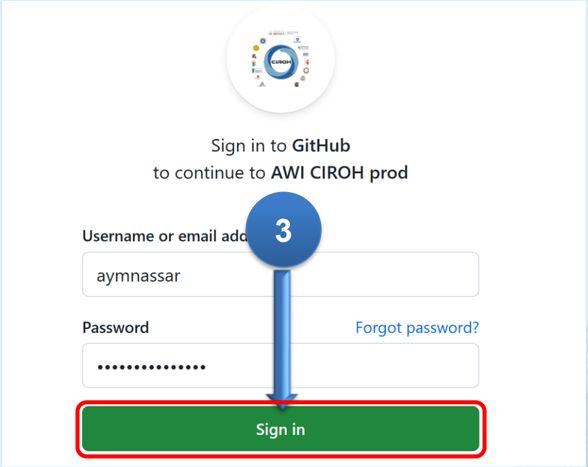
Container with

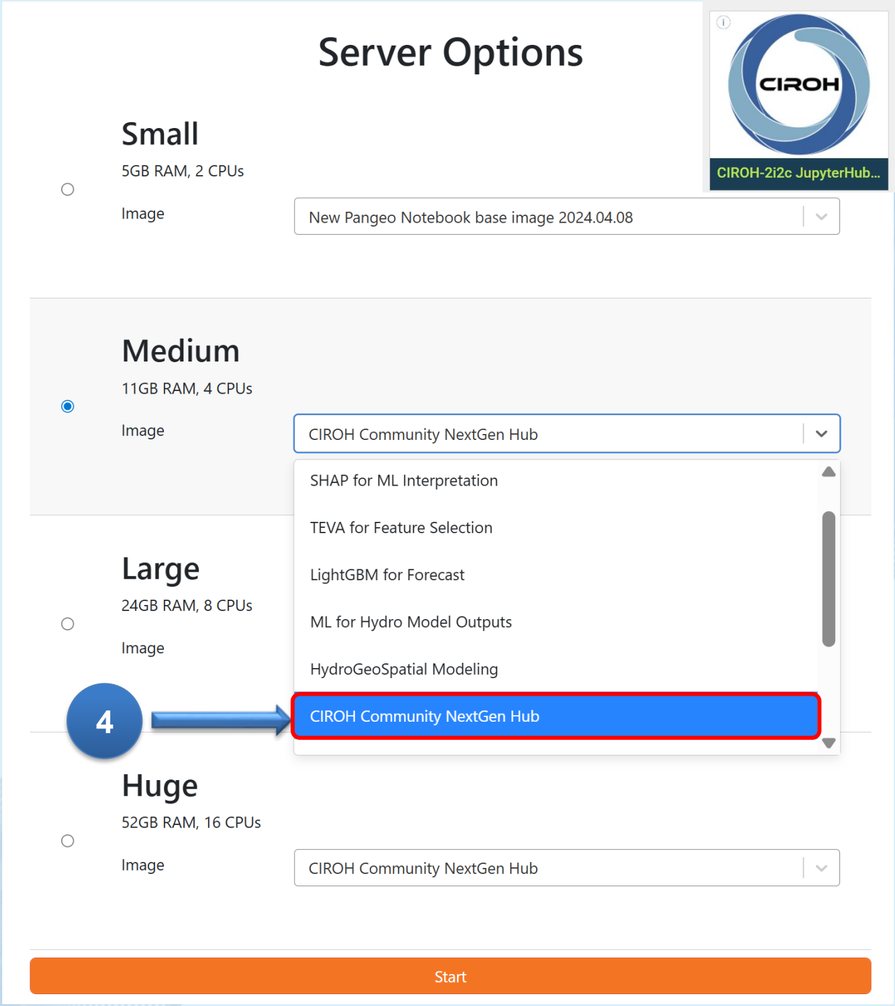
- NextGen In A Box (NGIAB) software for NextGen
- NextGEN data preprocess tools
- 2i2c JupyterHub
- Hsclient for interoperability with HydroShare repository
- Nbfetch for retrieving resources from HydroShare
- Hs_files for saving results back to HydroShare
- NextGEN community cyberinfrastructure ecosystem tools (e.g. TEEHR, Spotpy)



HydroShare-2i2c-JupyterHub application







NextGen Modeling Workflow Notebooks

1 NextGen Data Preparation

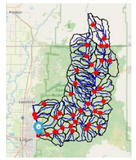
Authors:

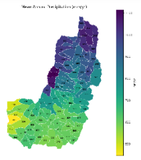
- Ayman Nassar (ayman.nassar@usu.edu)
- David Tarboton (david.tarboton@usu.edu)
- Furqan Baig (fbaig@illinois.edu)

Hydrofabric

Forcings

Configurations





Last Updated: 10/27/2025

Purpose:

This notebook simplifies the preparation of input data for running NextGen. It helps users subset and retrieve hydrofabric and forcing data, as well as set up model configurations and realizations. For more information on NextGen data preparation, refer to this **link**.

Audience:

Researchers and graduate students who are familiar with Jupyter Notebooks, basic Python, and basic hydrologic data analysis.

Description:

This notebook accepts inputs such as a gage ID, catchment ID, or Vector Processing Unit (VPU) to define a specific location, along with start and end dates for the time period of interest. It utilizes geometry and model attributes from the **Hydrofabric Dataset**. It uses AORC forcing data at a 1-km grid resolution. The forcings are computed as a weighted mean of the gridded AORC data, with weights derived using the `Exact Extract` tool and calculated using `NumPy`.

2 NextGen Model Implementation/Run

Authors:

- Ayman Nassar (ayman.nassar@usu.edu)
- David Tarboton (david.tarboton@usu.edu)
- Furqan Baig (fbaig@illinois.edu)

Python wrapper (pyNGIAB)

Last Updated: 11/15/2025

Purpose:

This Jupyter Notebook enables the execution of the NextGen using a **Python wrapper**. The input data, subset for the Area of Interest (AOI), is prepared using the separate notebook titled "NextGen Data Preparation" and is utilized here to drive the model simulation.

Audience:

Researchers who are familiar with Jupyter Notebooks, basic Python, and basic hydrologic data analysis.

Description:

The **Python wrapper** for running NextGen allows seamless model execution directly within the Jupyter environment, eliminating the need for terminal commands.

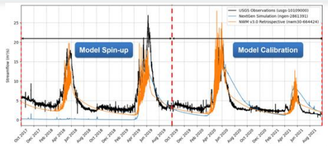
Data Description:

The user must provide the path to the preprocessed data generated using the "NextGen Data Preparation" Jupyter notebook. This directory should include the hydrofabric subset, meteorological forcings, and all necessary model configuration and realization files.

3 TEEHR for NextGen Model Output Analysis and Evaluation

Authors:

- Ayman Nassar (ayman.nassar@usu.edu)
- David Tarboton (david.tarboton@usu.edu)



Last Updated: 12/01/2025

Purpose:

TEEHR (**T**ools for **E**xploratory **E**valuation in **H**ydrologic **R**esearch) is a Python package that provides a flexible framework for evaluating hydrologic model performance. In this notebook, **TEEHR** is used to assess NextGen model outputs by comparing simulated streamflow with `USGS observations` and the `National Water Model Retrospective Analysis v3.0`. The workflow demonstrates how to set up the TEEHR evaluation environment, organize required datasets, and generate performance metrics and diagnostic plots. More information about **TEEHR** can be found in the **TEEHR User Guide**.

Audience:

Researchers, hydrologists, practitioners, and graduate students working with NextGen hydrologic simulations. Users should be familiar with Python, Jupyter Notebooks, and basic hydrologic modeling concepts.

Description:

This notebook leverages **TEEHR** to load, organize, and evaluate hydrologic datasets needed to assess NextGen streamflow simulations.
It performs the following tasks:

- Reads `NextGen-simulated streamflow` outputs.
- Aligns them with `USGS observed discharge` and `NWM v3.0 retrospective simulations`.
- Sets up or clones a `TEEHR evaluation directory`.
- Computes performance metrics and generates diagnostic plots for model evaluation.

4 Data Analysis and and Visualization of NextGen Outputs

Authors:

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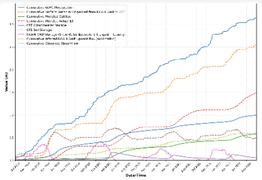
- David Tarboton (david.tarboton@usu.edu)
- Furqan Baig (fbaig@illinois.edu)

Water Balance Analysis



Last Updated: 11/16/2025

Purpose:

This notebook provides a workflow for **processing, and visualizing NextGen model outputs**. It enables users to examine divide-level (subcatchment) variables, compute watershed-level weighted averages, and generate diagnostic plots to better understand model behavior. It is designed as a companion to the *NextGen Data Preparation* notebook and focuses on analyzing model outputs produced after a successful NextGen simulation.

Audience:

Researchers, hydrologists, practitioners, and graduate students working with NextGen hydrologic simulations. Users should be familiar with Python, Jupyter Notebooks, and basic hydrologic modeling concepts.

Description:

This notebook requires a directory containing **NextGen simulation outputs**, including variable outputs for each divide (subcatchment), flowpath routing, and other model results. It utilizes geometry and model attributes from the **HydroFabric Dataset** to spatially link divide IDs with their corresponding attributes. Weighted watershed averages are computed using the area of each divide to ensure that aggregated time series accurately represent the full watershed behavior.

Utility functions from `hydrofabric_visualization_utils.py` and `ngen_outputs_utilis.py` support the extraction, processing, and visualization of model outputs. This includes generating interactive maps, subcatchment-level variable plots, and aggregated watershed-level hydrographs.

5 NextGen Model Calibration

Authors:

- Ayman Nassar (ayman.nassar@usu.edu)
- Joshua Cunningham (jcunningham8@ua.edu)
- David Tarboton (david.tarboton@usu.edu)

SPOTPY Calibration

Last Updated: 10/26/2025

Purpose:

This notebook is designed to automate the calibration workflow for NextGen streamflow. It focuses on calibrating key parameters of the **NOAH-OWP** and **Conceptual Functional Equivalent (CFE)** hydrologic models to improve the agreement between simulated and observed streamflow.

Audience:

Researchers and graduate students who are familiar with Jupyter Notebooks, basic Python, calibration, and basic hydrologic data analysis.

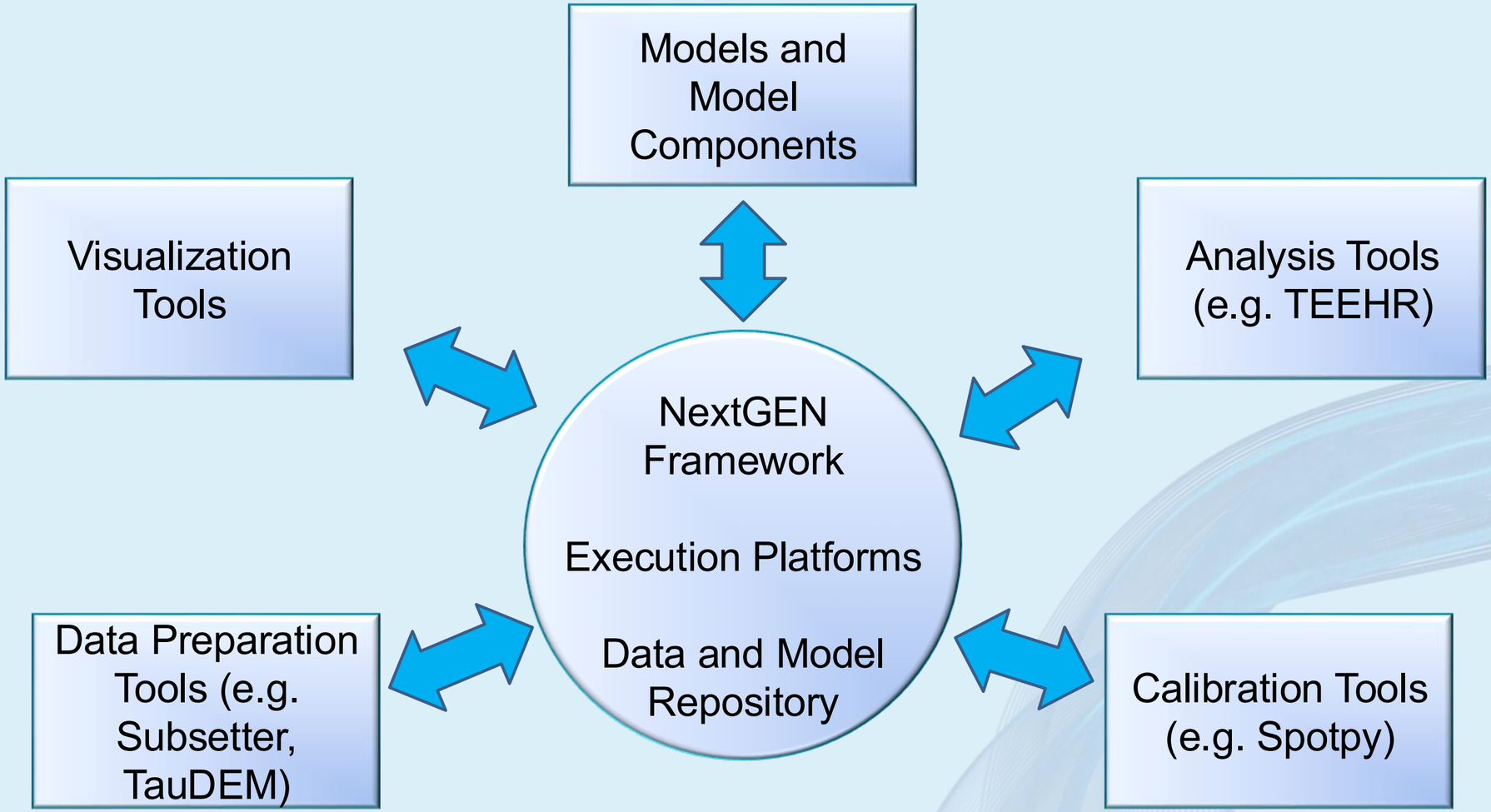
Description:

The calibration process uses SPOTPY (https://github.com/thouska/spotpy) to optimize model parameters and improve the match between simulated and observed streamflow. In this workflow, you can select either the **DDS** (Dynamically Dimensioned Search) or **SCE-UA** (Shuffled Complex Evolution) algorithm to search for the best parameter values. You can also use **Kling–Gupta Efficiency (KGE)** or **Root Mean Square Error (RMSE)** as the performance measure to evaluate how well the model reproduces observed flow.

The workflow starts by defining the **simulation period**, the **gage ID**, and the **feature ID**, then downloading observed streamflow data from the U.S. Geological Survey (USGS). SPOTPY then begins the calibration by selecting an initial set of parameters for the **NOAH-OWP** and **Conceptual Functional Equivalent (CFE)** models from predefined parameter ranges. These parameters are written into the NextGen configuration, and the model is run to generate simulated streamflow.

The simulated results are then compared with the observed data using the chosen performance measure (**KGE** or **RMSE**). Based on how well the model performs, the optimization algorithm adjusts the parameters and runs the model again. This cycle of **updating parameters**, **running the model**, and **evaluating performance** continues over many iterations. Gradually, the parameters are improved, resulting in a final calibrated model that provides the best match between simulated and observed streamflow for the selected gage.

A software and collaboration ecosystem where any one researcher can focus on their contribution in one “spoke” while using common or established “best practice” tools of others



General approach to web-based collaboration environment for collaborative and FAIR aligned modeling

- Sharing and publication of data (DOI)
- Social discovery and added value
- Model and workflow sharing

Repository

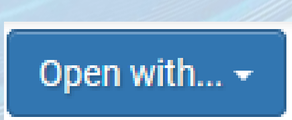
Collaboration, Reproducibility, Credit, Transparency

- Model input data preparation
- Model execution
- Visualization and analysis (best of practice tools)

Linked Computing

Server/Cloud Computation

- Platform independence
- Big data
- Reproducibility
- Interoperability
- Reduced needs for software installation and configuration



Conclusions

- NextGen complexity and reliance on advanced cyberinfrastructure (CI) present significant challenges for researchers and graduate students
- Lowering technical barriers is critical for enabling collaborative hydrologic research with broader community participation
- Building on tools from “NextGen In A Box (NGIAB)” we configured the CIROH-2i2c JupyterHub environment to support NextGen and its data preparation tools
- This serves as a foundation for community research to improve NextGen models
- Improved tools for user to save results to HydroShare repository

This research was supported by the Cooperative Institute for Research to Operations in Hydrology (CIROH) with funding under award NA22NWS4320003 from the NOAA Cooperative Institute Program. The statements, findings, conclusions, and recommendations are those of the author(s) and do not necessarily reflect the opinions of NOAA.

CIROH

Thank you

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HydroShare Resource